

Computational Complexity Analysis

To provide a rigorous quantitative comparison between the proposed DMA3C and the conventional MADDPG baseline, this section derives the parameter counts for their respective critic networks within the 3375-node distribution system.

1) *MADDPG parameter analysis*: In the centralized training and decentralized execution (CTDE) framework, the global critic processes the joint state and action information from all n agents. For the 3375-node system, the global state dimension is 2818 and the global action space dimension is 432. Consequently, the centralized critic must process a joint state-action input of 3250 dimensions. Given a centralized multi-layer perceptron (MLP) architecture with hidden layers of (512, 128), the total number of parameters P_{MADDPG} is calculated as:

$$P_{\text{MADDPG}} = (3250 \times 512 + 512) + (512 \times 128 + 128) + (128 \times 1 + 1) = 1730305 \approx 1.73 \times 10^6$$

2) *DMA3C parameter analysis*: In contrast, the proposed DMA3C architecture leverages the sparse spatial coupling inherent in distribution systems. Each critic network represents a truncated value function $V_i(s_{N_i^\kappa})$ for agent i . Unlike the centralized critic, this localized function depends only on the state $s_{N_i^\kappa}$ within the κ -hop neighborhood of agent i rather than the global state. In this test system, the maximum input dimension for the truncated value function is reduced to 592. Under a (256, 128) architecture, the parameter count P_{DMA3C} for the most complex local agent is calculated Appendix-B-Computational Complexity Analysis as:

$$P_{\text{DMA3C}} = (592 \times 256 + 256) + (256 \times 128 + 128) + (128 \times 1 + 1) = 184833 \approx 1.85 \times 10^5$$

3) *Comparative summary*: The comparative analysis reveals that the parameter count of the proposed DMA3C critic is approximately one order of magnitude lower than that of the MADDPG baseline. By decentralizing the training process and truncating the state space, the proposed method effectively mitigates the curse of dimensionality. This structural advantage ensures superior computational efficiency and facilitates stable convergence in large-scale DSR tasks.